Maximize Afternoon Coffee Profitability with Python

After sitting down with Alex, the owner of Afternoon Coffee in Gainesville, I realized he was trapped in a classic "Founder’s Dilemma." He was working incredible hours, but the bank account wasn't reflecting the effort.

During our conversation, it became clear that he was guessing on inventory and feeling completely overwhelmed by the weekend rush. I decided to step in as a consultant and use Linear Programming to turn his qualitative stress into a quantitative roadmap for profit.

The Story of Afternoon Coffee

I started by identifying the specific environment Alex operates in. During the week, it's a steady flow of students and regulars, but every Saturday at 9 AM, the local run club "raids" the shop. Alex mentioned that he often has to cater to 2x or 3x his usual demand during this window.

Problem Statement

I wanted to help Alex maximize his daily profit while solving three distinct problems he shared with me:

- Labor Capacity: Alex and his team have a fixed number of minutes to handle orders, prep food, and steam milk before the line gets too long.
- The "Two-Minute" Rule: Alex observed that if customers wait more than a couple of minutes, they lose interest and might not return.
- Inventory Waste: He noticed that on weekdays, items were consistently left over after closing at 3 PM, cutting into his margins.

Methodology: Building the Model

I built a PuLP model in Python to find the "Sweet Spot" for his operations. I defined his menu items not just by price, but by their "Labor Cost" in minutes.

- Available Resources (Weekdays): I modeled Alex plus one employee for the 3-hour peak, totaling 360 minutes.
- Available Resources (Saturdays): I started with his current team of three for the 3-hour "Run Club" surge, totaling 540 minutes.

The Weekday Labor Question

Alex was specifically unsure whether he should hire a third person for the 9 AM to 12 PM weekday shift to help with cleaning and stocking while the rush is on. After interviewing Alex, I ran the numbers through our model to see if this "help" translated into profit.

The results were eye-opening:

- Current Weekday Profit: $426.50 (with 2 people).
- Profit with 3rd Person: $426.50.
- Net Profit Impact: -$45.00 (due to the $15/hr labor cost).

Alex mentioned that the rush dies down after 12 PM, and the data confirms it. Because demand is naturally capped on weekdays, a third person doesn't unlock more sales; they just create a higher expense.

My recommendation was clear: do not hire a 3rd person for weekday shifts.

Alex can manage the "overwhelmed" feeling by using the 37 minutes of "idle time" our model identified for operational tasks like cleaning.

Results: Solving the Saturday Surge

When I ran the first simulation of the Saturday rush, I saw that with only three people, Alex was leaving revenue on the table.

The model was forced to pick "faster" items like basic Bacon & Cheese sandwiches just to keep the line moving, sacrificing the high-margin "Egg, Bacon, and Cheese" options runners actually wanted.

Profit reached with 3 People: $1,090.00

I then tested a scenario where Alex hires a 4th person for that Saturday morning shift. The shift in results was massive:

- Specialty Drink Volume: Increased from 50 to 90 units.
- High-Margin Sandwiches: Jumped from 2 units to 60 units.
- Net Profit Gain: $116.00 extra every Saturday (after paying the $45 labor cost).

1	Afternoon Coffee Data			
2				
3	Item	Category	Profit ($)	Labor (Min)
4	**Drip/Cold Brew**	Drink	3.3	1
5	**Specialty (Latte/Matcha)**	Drink	3.65	4
6	**Bacon & Cheese (Sandwich)**	Food	4.75	3
7	**Egg, Bacon & Cheese (Sandwich)**	Food	5.5	4
8	**Pastries**	Food	3	0.5

Tackling Weekday Perishability

Next, I addressed the leftover issue Alex mentioned. I implemented a Perishability Ceiling in the model to stop him from over-preparing items that usually end up as waste, while a Loyalty Floor ensures he always has enough for his regulars.

REFINED WEEKDAY SWEET SPOT

- Drip: 50.0
- Specialty: 30.0
- Sandwiches: 20.0
- Pastry: 20.0
  Labor Used: 248 / 360 mins (Giving Alex a 20% "Breather Buffer" for the operations he was worried about).

The Gainesville Weather Edge Case

To help Alex navigate the "Gainesville Weather Edge Case," I had to address a fundamental risk: Inventory Obsolescence caused by external variables.

In Gainesville, a bright morning can turn into a torrential downpour in minutes, which Alex noted often results in a "ghost town" effect at the shop even after he has fully prepped for a 60-person run club surge.

The Problem with Fixed Batching

Alex’s current method involved "guessing" and pre-preparing a large batch of sandwiches early in the morning to handle the expected 9 AM raid.

- The Waste Risk: If the rain hits at 8:45 AM, the runners don't show up, and Alex is left with 60 pre-heated or pre-assembled sandwiches that have a shelf life of just a few hours.
- The Revenue Loss: Because these items are highly perishable, they cannot be resold on Sunday, leading to a total loss of both ingredient costs and the labor time spent preparing them.

The Hybrid Prep Strategy: "Just-in-Time" Assembly

My solution for Alex was to leverage the extra labor capacity we identified in the 4-person Saturday model to shift from "Fixed Batching" to "Just-in-Time" (JIT) assembly.

- Component Pre-Prep: During the early shift, the team preps only the non-perishable components (e.g., slicing bread, portioning cold cheese and bacon).

- The "Clear Skies" Trigger: Alex monitors the weather radar; if it is clear at 8:30 AM, the 4th employee begins "Stage 2" assembly heating and final wrapping just minutes before the club arrives.
- On-Demand Surge: If the rain holds off but the crowd is smaller than expected, the extra employee allows the shop to switch to 100% made-to-order service.

Why This Works for Alex

This strategy effectively turns a Fixed Cost (the prepped food) into a Variable Cost that only triggers when demand is certain.

- Waste Mitigation: By not pre-heating the sandwiches, the ingredients stay fresh in the fridge and can be used for the Sunday shift or Monday weekday rush if the rain cancels Saturday's event.
- Labor Efficiency: The 4th person we recommended for Saturdays acts as the "Insurance Policy." They provide the speed needed to assemble items quickly on-demand, ensuring Alex still hits his "Two-Minute" delivery KPI even without a pre-made pile of food.

This approach transforms the weather from a business-ending threat into a manageable operational variable.

Final Consultation Verdict

I sat down with Alex to review the final results of the PuLP model. My goal was to transform his "gut feelings" and operational stress into undeniable mathematical truths.

The data proved that his sensation of being overwhelmed on Saturdays was not just in his head. It was a direct result of a labor bottleneck that was actively strangling his profit.

The Quantitative Truth Behind the Saturday Rush

I discovered that with only three people during the Saturday "Run Club" surge, the shop reaches a point of total labor saturation. The model showed that in this "survival mode," Alex's team is forced to prioritize speed over margin.

By adding a fourth person, we aren't just adding a body to the floor. We are fundamentally shifting the shop's production curve to allow for high-margin items like Specialty Matchas and Egg, Bacon, and Cheese sandwiches.

The Weekday "False Alarm" on Labor

On the flip side, I proved to Alex that hiring more help during the week would be a financial mistake. Because the weekday demand is naturally capped by local Gainesville foot traffic, an extra employee becomes a sunk cost that cannot be recovered.

The stress Alex feels during the week actually stems from a "Production Imbalance" rather than a lack of hands. I showed him that by using the model's "Sweet Spot" prep list, he can fulfill all orders while still maintaining a 20% breather buffer.

Protecting the Profit from the "Trash Can"

Alex mentioned he was constantly throwing away food at 3 PM on weekdays. I built the "Perishability Ceiling" into the model to specifically target this issue.

This constraint dictates exactly how much to prepare so that the last sandwich is sold just as the doors close. By adhering to these "Sweet Spot" numbers, we effectively protect his margins from being tossed into the trash.

Closing the Gap on Uncertainty

Finally, I addressed the "Gainesville Rain" problem that Alex was so worried about. I proved that with the correct labor structure on Saturdays, he doesn't need to risk pre-batching 60 units of food.

The data is clear: hiring that extra person on Saturday pays for itself nearly three times over. Conversely, the weekday strategy focuses on lean prep, ensuring the shop remains profitable and stress-free regardless of the weather.